Konrad Yee

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BASc Civil Engineering + PEY Co-op, University of Toronto | Available May 2026

EDUCATION

Bachelor of Applied Science — Civil Engineering + PEY Co-op | University of Toronto *Sept 2024 – Apr 2029 (Expected)*
Relevant Coursework: Construction Management (Revit), Engineering Strategies & Practices (Excel/Gantt), Statics

WORK EXPERIENCE

Bridge Design Specialist | Engineers in Action — University of Toronto Chapter | Toronto, ON *Sept 2025 – Present*

- Co-led structural design of a 65.5 m suspended pedestrian bridge in Ácora, Peru, serving ~300 residents producing full AutoCAD construction drawings, calculation packages, and contract-ready documentation from site constraints.
- Engineered optimized abutment configurations (2G-80A left, 1G-80A right) to minimize excavation volume while fully satisfying EIA setback requirements and slope constraints, reducing projected construction cost and site disruption.
- Performed cable tension and sag analysis per the 2025 EIA Bridge Binder (Vol.2), achieving a Factor of Safety of 3.46 against the 3.0 requirement and refining sag geometry to raise flood freeboard to 6.81 m, improving long-term infrastructure resilience.
- Prepared a comprehensive Bill of Quantities (BOQ) and coordinated submissions with Bridge Corps engineers, DEIC/FEIC reviewers, faculty advisors, and local community partners to achieve full regulatory approval.
- Verified construction-stage hoisting feasibility at 3% cable sag to confirm anchor and winch system capacity, ensuring the build sequence was executable under field conditions.

Structural Engineering Intern | Exactus Energy | Toronto, ON *Jun 2025 – Aug 2025*

- Delivered structural assessments and code-compliance documentation for 60+ rooftop solar installation projects across residential civil systems, meeting all client deadlines independently across a full 3-month term.
- Calculated wind, seismic, and uplift loads per ASCE, NBC, IBC, and IEBC standards to size framing members and verify connection capacities, directly supporting permitting and construction cost estimation packages.
- Modelled framing systems and validated structural load paths using WoodWorks, AutoCAD, BricsCAD, and Excel, reducing design revision cycles by catching structural deficiencies earlier in the review process.
- Led roofing damage investigations and retrofit feasibility studies for diverse project types, delivering prioritized repair and structural reinforcement recommendations that informed client decision-making.
- Applied QGIS for site-condition validation and maintained comprehensive project records in Dynamics CRM and Smartsheet, improving documentation traceability across multi-disciplinary engineering teams.

STEM Instructor | Focus on Youth Day Camp | Etobicoke, ON *Jul 2023 – Aug 2023*

- Designed and delivered 3+ original hands-on STEM modules from scratch, adapting instruction to individual skill levels across groups of 15–20 campers per session to maximize engagement and concept retention.
- Elevated program quality by mentoring 2 junior instructors on lesson delivery technique and helping standardize cross-disciplinary curriculum materials used across the full instructor team.
- Maintained a 100% incident-free record across all sessions by enforcing structured safety protocols and proactively identifying and resolving hazards before the start of each activity.

Engineering Intern — Civil Infrastructure | Green Infrastructure Partners Inc. | North York, ON *Mar 2023 – Jun 2023*

- Reduced data retrieval time by 50% by designing and deploying an Excel-based asphalt pile inventory and reference system from scratch, replacing fragmented tracking with a single centralized database.
- Integrated Propeller drone software and QGIS to replace manual pile estimation with precision volumetric calculations, increasing inventory measurement accuracy and supporting more reliable procurement decisions.
- Authored standardized materials handling and equipment-use guides for field workers, cutting new-hire onboarding time by 10% and establishing consistent operational workflows across the site team.
- Performed quantity take-offs and aggregate demand calculations for municipal infrastructure projects, producing analytical reports that directly informed construction cost estimates and procurement planning.

Facilities Management Assistant | Black Creek Co-operative Homes | Toronto, ON *Jun 2022 – Aug 2022*

- Resolved 50+ maintenance work orders on time by coordinating directly with contractors and external vendors, tracking all tasks from assignment to sign-off to ensure timely completion within committed service windows.
- Reduced average maintenance response time by 13% by redesigning the inspection and reporting workflow, introducing a priority-tiered system that fast-tracked critical repairs and eliminated backlog bottlenecks.
- Lowered contractor dependency and operating costs by personally executing hands-on carpentry, plumbing, and electrical repairs, reducing the volume of tasks requiring external vendors.
- Improved long-term asset reliability by implementing structured preventive inspection protocols that identified and escalated urgent repair needs before they escalated into larger deficiencies.

PROJECTS & CO-CURRICULAR

Motion Commotion Technical Transportation Debate — 3rd Place *Feb 2026*

- Placed 3rd in a competitive technical transportation debate at the ITE University of Toronto chapter, demonstrating applied knowledge of civil infrastructure systems and professional engineering communication skills.

Structural Project Lead — Modular Cart System | AutoCAD, Rhino, Excel, Microsoft Office *Jan 2025 – Apr 2025*
University of St. Michael's College, Toronto, ON

- Designed and delivered a modular mobility and bed-stop system for a mid-century etching press using AutoCAD, enabling safe single-operator use and satisfying all client functional, safety, and accessibility requirements.
- Led a cross-functional team of 6 through structured ideation of 68+ design concepts, applying morphological charts and Pugh matrices in Excel to systematically converge on the optimal solution.
- Confirmed a 1,400 lb load capacity through detailed force modelling and load-path verification in AutoCAD, significantly improving press operational safety without compromising print quality.
- Drove on-time delivery of all project milestones by managing schedules, stakeholder communications, and team coordination using Microsoft Office throughout tight academic deadlines.

Cardboard Structural Chair Design — "Ampersand" | AutoCAD, Physical Prototyping *Sept 2023 – Jan 2024*

- Engineered and fabricated a full-scale load-bearing chair from corrugated cardboard with zero adhesives or fasteners — achieving a single-load test capacity of 220 lbs and a sustained 150 lb hold for 30 minutes.
- Developed rapid paper prototypes and an AutoCAD model to optimize structural geometry and minimize material waste, completing the final build using only two 48" x 96" sheets with near-zero offcuts.
- Designed interlocking and vertical locking joint systems with deliberate grain orientation to maximize load-path efficiency, eliminating the need for any mechanical fastening while maintaining structural integrity.
- Documented construction tolerancing and alignment solutions to ensure full design reproducibility, demonstrating disciplined approach to fabrication quality control.

Warren Truss Bridge Design & Structural Analysis | AutoCAD, CAD Structural Analysis *Sept – Dec 2022*

- Designed and fabricated a 70 cm arched Warren truss bridge from 70 popsicle sticks, applying CAD-based structural analysis to optimize member geometry for strength, constructability, and cost efficiency, achieving a load-to-weight efficiency ratio of 234.6.
- Simulated a $172,000 full-scale cost model and scaled the digital design by 2.5x to produce fabrication-ready member lengths, directly applying structural analysis outputs to construction cost estimation.
- Reinforced critical stress points through lamination and cross-bracing strategies, then load-tested the completed structure to a 30 lb capacity at 58 g total weight.
- Conducted post-test failure mode analysis and proposed targeted design improvements to increase tensile performance in high-stress members, developing engineering judgment for iterative design.

TECHNICAL SKILLS

CAD & Design: AutoCAD, Civil 3D, MicroStation, BricsCAD, Revit, Rhino 3D, SketchUp, SolidWorks, Floor Planner

Engineering Analysis: WoodWorks®, MATLAB, structural load analysis (wind, seismic, uplift), quantity take-offs, cost estimation, force modelling

GIS & Spatial: ArcGIS, QGIS, Propeller® (drone volumetrics)

Project & Data Tools: Microsoft Office (Excel, Word, PowerPoint), Power BI, Smartsheet, Dynamics CRM, Notion, Slack, Teams, Google Suite, Jupyter Notebook

Programming: Python, C, MATLAB

Core Competencies: Structural analysis, constructability review, technical documentation, construction cost estimation, quantity take-offs, materials management, project coordination, multi-disciplinary collaboration